# Aarush Singh

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## Education

**University of Waterloo** — Waterloo, ON
*Bachelor of Mathematics in Computer Science (Honours) – Presidential Scholarship* — *Sep. 2025 – Sep. 2030*

**Wilfrid Laurier University** — Waterloo, ON
*Bachelor of Business Administration (Honours)* — *Sep. 2025 – Sep. 2030*

**Relevant Coursework:** Accounting, Finance, Organizational Behaviour

## Skills & Certifications

**Certifications:** Bloomberg Market Concepts (BMC), Bloomberg Finance Fundamentals
**Business Tools:** Microsoft Excel, Word, PowerPoint, Tableau, SharePoint
**Languages:** Python, SQL, C++, C, JavaScript, TypeScript
**Data & ML:** pandas, NumPy, scikit-learn, PyTorch, Matplotlib
**Cloud & Databases:** PostgreSQL, Supabase, AWS (Lambda, S3, Aurora, Bedrock), Git

## Experience

**Research Analyst – FICC** — Waterloo, ON
*Whitestone Capital* — *Oct. 2026 – Present*
- Conduct fundamental research on **rates, currencies, and commodities** for a cross-university equity research firm.
- Develop thesis-driven, investment-grade research through industry analysis and **financial modeling** across FICC markets.

**Software Engineering Intern** — New Delhi, IN
*Ministry of Statistics and Programme Implementation* — *May 2026 – Aug 2026*
- Built and maintained databases with SQL, Python, and Excel that now hold more than **10.8 million records**.
- Designed Python ETL pipelines loading **80+ survey workbooks** (Irrigation, ASUSE, Gender) into the Unified Data Platform.
- Contributed to this flagship Government of India platform, which publishes more than **100,000 datasets**.
- Benchmarked AI data processing tools on three flagship national datasets (NSS, NAS, AISHE) to improve reliability.

**President** — Waterloo, ON
*Movie Club, University of Waterloo* — *Jan. 2026 – Apr. 2026*
- Organized recurring film screenings, handling venue booking, film selection, promotion, and event-day logistics.
- Managed the club's **treasury and budget**, tracking spending and allocating funds across screenings and events.
- Led the executive team and grew attendance by **30–40% per screening** through targeted outreach.

**BDO Future Leaders Challenge – Semi-Finalist** — Waterloo, ON
*BU121, Wilfrid Laurier University* — *Mar. 2026*
- Selected among the **top 17%** of teams to present SDG-focused solutions for enterprise firms.
- Built a **data governance framework** helping geophysical service firms keep pace with rising compliance demands.

**Director of First Year Affairs** — Waterloo, ON
*Mathematics Society, University of Waterloo* — *Sep. 2025 – Dec. 2025*
- Coordinated **4 elected representatives** and approved internal communications reaching **5,000+ Math students**.
- Chaired the representative committee and ran the Numerica quiz contest for **50+ first-year participants**.

## Projects

**Volatility Forecasting Tool** | *Python, PyTorch, scikit-learn, pandas* — 2026
- Forecast next-week **S&P 500 volatility** with **EWMA**, **HAR**, and **LSTM** models trained on **16+ years** of daily data.
- The LSTM cut forecast error (RMSE) by **12%** vs. EWMA and **6%** vs. HAR on out-of-sample 2022–2026 data.
- Built a **leakage-free pipeline** and a **CLI tool** that retrains weekly, forecasts any ticker, and tracks its own accuracy.

**Brokeberg: Political Intelligence Terminal** | *Python, AWS, PostgreSQL/pgvector, Bedrock, Next.js* — 2026
- Built an **event-driven ETL pipeline** on **AWS Lambda, EventBridge, S3, and Aurora** ingesting **7 public data sources**.
- Built a **6-pass LLM extraction pipeline** on **Bedrock** that ties each claim to source text, using Haiku or Sonnet per pass.
- Linked **economic indicators** to politicians and competitive Senate races in a Bloomberg-style **terminal UI**.

**Campus Marketplace Web Application** | *Next.js 15, Supabase (PostgreSQL, Auth, Storage), Tailwind CSS* — 2026
- Built a full-stack marketplace for **@uwaterloo.ca** accounts with listings, wanted posts, filters, and sortable search.
- Designed **11 PostgreSQL tables** with **53 row-level security policies** and **27 indexes** for role-based access.
- Built an **admin dashboard** for content moderation and onboarded **50+ early-adopter students** to the platform.